$$1+\frac{y^2}{q^2} = cq$$

$$q^2+y^2 = c\,q^3 \qquad (2)$$

Given output is 1 unit and revenue is ₹5

$\therefore$ (2) $\Rightarrow$ $1+25=c \Rightarrow c=26$

$\therefore$ The total revenue fonction is

$q^2+y^2=26q^3$

Exercise 4.3

Solve the following homogeneous differential equations.

1. $x\dfrac{dy}{dx}=x+y$
2. $(x-y)\dfrac{dy}{dx}=x+3y$
3. $x\dfrac{dy}{dx}-y=\sqrt{x^2+y^2}$
4. $\dfrac{dy}{dx}=\dfrac{3x-2y}{2x-3y}$
5. $(y^2-2xy)dx=(x^2-2xy)dy$
6. The slope of the tangent to a curve at any point (x, y) on it is given by $(y^3-2yx^2)dx+(2xy^2-x^3)\,dy=0$ and the curve passes through (1, 2). Find the equation of the curve.
7. An electric manufacturing company makes small household switches. The company estimates the marginal revenue function for these switches to be $(x^2+y^2)dy=xydx$ where x represents the number of units (in thousands). What is the total revenue function?

4.2.4 Linear differential equations of first order:

A differential equation is said to be linear when the dependent variable and its derivatives occur only in the first degree and no product of these occur.

The most general form of a linear equation of the first order is $\dfrac{dy}{dx}+Py=Q$ (1)

P and Q are functions of x alone.

Equation (1) is linear in y. The solution is given by $ye^{\int pdx}=\int Qe^{\int pdx}dx+c$. Here $e^{\int pdx}$ is known as an integrating factor and is denoted by I.F.

Note

For the differential equation $\dfrac{dx}{dy}+Px=Q$ (linear in x) where P and Q are functions of y alone, the solution is $xe^{\int pdy}=\int Qe^{\int pdy}dy+c$

Example 4.20

Solve $\dfrac{dy}{dx}+\dfrac{y}{x}=x^3$

Solution:

Given $\dfrac{dy}{dx}+\dfrac{1}{x}y=x^3$

It is of the form $\dfrac{dy}{dx}+Py=Q$

Here $P=\dfrac{1}{x}$, $Q=x^3$

$$\int Pdx = \int\frac{1}{x}dx=\log x$$

$$\text{I.F} = e^{\int pdx}=e^{\log x}=x$$

The required solution is

$$y(I.F) = \int Q(I.F)dx+c$$

$$yx = \int x^3.x\,dx+c$$

$$= \int x^4dx+c$$

$$= \frac{x^5}{5}+c$$

$$\therefore\ yx = \frac{x^5}{5}+c$$

XII Std - Business Maths & Stat EM Chapter 4.indd 92 2/28/2020 6:02:28 PM